

2.7 Proofs – Part 1

Lesson Introduction

 Benchmarks	MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.		 Learning Targets	- I can relate deductive reasoning to completing proofs based on an assumption or given. - I can recognize paragraph, flowchart, and two column proofs.	
 Benchmark Coherence	Building On MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.		Working Toward N/A		
 Essential Questions	- How can I apply what I've learned about deductive reasoning to complete proofs? - What are the differences between the three different proof types?				
 Lesson Narrative	This lesson is the first of a two-part lesson series on proofs. In this lesson, students connect their learning from the prior lessons to begin formally writing proofs. They will be introduced to three formal types of proofs: paragraph proofs, two-column proofs, and flowchart proofs. Students will compare the three types of proofs presented for the same given and conclusion.				
 Component Summary	2.7.1 Warm-Up (~5 min.)	Observing symmetry in Art Deco (<i>SE p. 112</i>)	2.7.3 Guided Instruction (10 min.)	Completing a flowchart proof (<i>SE p. 115</i>)	
	2.7.2 Exploration (20 min.)	Connecting conditionals to formal proof writing (<i>SE p. 112</i>)	2.7.4 Guided Practice (10 min.)	Completing a two-column proof (<i>SE p. 115</i>)	
	2.7.5 Wrap-Up (~5 min.)	Four quadrant self-reflection on the interest and understanding of this lesson (<i>SE p. 116</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Paragraph Proof - Two-column Proof - Definition of Congruence - Flowchart Proof <u>Additional Glossary:</u> N/A		N/A		- This lesson is the first of a two-lesson series on proofs. Be sure to work through both lessons prior to beginning this one so that you know where the series will end. This can help with inadvertently adding too much information in this lesson by itself. - Consider creating an anchor chart to capture the three different proof formats for the same proof. This will help students get more comfortable with each format as well as with the steps in the process.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may forget to use a given statement as part of their original proof. - Students may skip a step in the proof process. - Students may get lost in the flow of a flowchart proof. - Students may make assumptions based on the diagram instead of using appropriate reasons.	- If one has not already been started, consider having the students create a proof resource booklet, notebook, or type of study guide that will continue to be added to throughout the course. This proof reference would include definitions, postulates, and theorems as well as possible hints (like typical next steps in proofs after a given statement). - Consider referring students back to their experience with the Uno card proofs and the logical flow of the steps. - For 2.7.2 Exploration consider whether your students need a more guided instruction. Take care that your decision does not remove productive struggle from your students or remove the opportunity for the student to engage in discussions with their peers. One possible consideration is to allow students to work independently for question 1 parts a – c and then work with their partner for question part d and beyond. Consider stopping the activity at each vocabulary term (such as paragraph proof) to review with the students and then release them to continue that section. For each section, review their answers to ensure they are on the right track with the outcomes of this activity.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share In 2.7.2 Exploration, students will work together to explore the concept of proofs. They will be set up to work on their own at first to have proper think time and then work with a partner to discuss their answers and move forward with additional exploration together.	MA.912.MTR.4.1: Discussion and Reflection The purpose of writing proofs is to provide a justification for why something is true. The act of writing formal proofs develops a student’s ability to present arguments based on evidence while also explaining methods and processes. Create a culture in which your students are comfortable asking questions to their peers and to you.
 Differentiate Instruction	In this lesson, students see the same proof presented in all three forms: two-column, paragraph, and flowchart. Consider using a visual aid, such as an anchor chart, to show the clear connection between the forms so that students can see the similarities.	
Lesson Notes:		

2.7.1 Warm-Up: Miami Art Deco

Component Overview: Students will observe Art Deco in the Miami area and answer questions related to symmetry.



2.7.1 Warm-Up: Miami Art Deco

A style known as Art Deco began in Paris in the mid-1920s. Art Deco buildings are characterized by their geometric shapes and symmetry. Two of the most famous examples of Art Deco architecture are the Empire State Building and Chrysler Building in New York City.

Photos of three locations in Miami that use this style of architectural design are shown.



1. Draw a line of symmetry on each picture.

Answers will vary. Sample answers on images above.

2. Which depiction of Art Deco do you find most interesting?

Answers will vary.



This activity is designed to connect art concepts to geometry concepts while highlighting Miami.



To add another element of engagement, consider asking students to find other pictures of art deco and describe their geometric shapes and symmetry.

2.7.2 Exploration: Proofs

Component Overview: Students will connect their prior learning from this unit to writing proofs.



2.7.2 Exploration: Proofs

A mathematical proof is an argument that shows something is true. A proof should begin with an assumption or a given statement.

1. A statement is given.

If two angles are supplementary to the same angle, then they are congruent to each other.

- a. What is the hypothesis?

Two angles are supplementary to the same angle.

- b. What is the conclusion?

The two angles are congruent to each other.

- c. Draw a picture for the if-then statement.



The hypothesis is the given, or what is known to be true.

The conclusion will need to be shown, or proven, using deductive reasoning.

- d. Discuss with your partner why inductive reasoning is not appropriate for this context. Summarize your discussion.

Answers will vary. Inductive reasoning is not appropriate because we are not looking at patterns and observations to draw conclusions. Instead, we are looking at facts and general principles.



Lessons 7 and 8 begin the more formal implementation of proofs. Students employ different types of proofs for the same given and prove. The repetitive nature focuses on learning the format of the different types of proofs. Also, by repeating the same thinking, the student is building memory for some common steps.



If students are not familiar with the vocabulary word “supplementary,” consider introducing the definition of supplementary angles prior to releasing students to question 1. If students are keeping a proof resource study guide of some type, have the students add this definition.

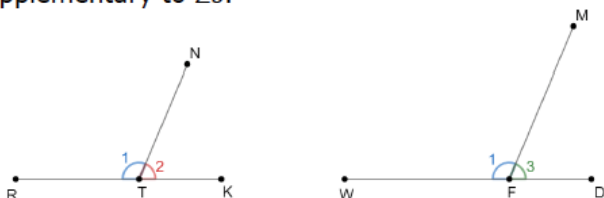
2.7.2 Exploration: Proofs (continued)

One way to show the deductive reasoning that's necessary to prove the conclusion is a paragraph proof.



A **paragraph proof** is a narrative that gives the steps with justifications that prove a statement as true.

In the diagram, $\angle 1$ is supplementary to $\angle 2$, and $\angle 1$ is supplementary to $\angle 3$.



2. Work with your partner to complete the paragraph proof.

It is given that $\angle 1$ is supplementary to $\angle 2$, and $\angle 1$ is supplementary to $\angle 3$. By definition of supplementary angles $m\angle 1 + m\angle 2 = 180^\circ$ and $m\angle 1 + m\angle 3 = 180^\circ$.

Since $m\angle 1 + m\angle 2 = 180^\circ$ and $180^\circ = m\angle 1 + m\angle 3$, then

$m\angle 1 + m\angle 2 = m\angle 1 + m\angle 3$ by the transitive property.

The result of applying the subtraction property of equality is

$m\angle 2 = m\angle 3$ By definition of congruence, $\angle 2 \cong \angle 3$.

Another type of proof is the two-column proof.



A **two-column proof** consists of two columns of a chronological list of steps, called statements, and justifications, called reasons.



Since each proof in this activity is the same concept but in a different format, some students may benefit from stations. Consider utilizing stations, consider having students with the strongest understanding begin with the paragraph proof. Students who are struggling should start with the two-column proof. If working with students who are struggling, consider using the two-column proof as a station where students work with you. Utilize the intended flexibility of this lesson to best suit the needs of your students. For students who have started at a proof other than the two-column proof utilize the time they spend with you to verify student answers and to address misconceptions before helping students with the two-column proof.



After completing questions 2 and 3, consider asking the following questions.
What do you think the role is of the given in each of the proofs? Why do you think that it is important to this process?



If students forget the definition of congruence as an option, refer students to Lesson 1.

2.7.2 Exploration: Proofs (continued)

3. Complete the two-column proof.

Statement	Reason
1. $\angle 1$ is supplementary to $\angle 2$, $\angle 1$ is supplementary to $\angle 3$	1. Given
2. $m\angle 1 + m\angle 2 = 180^\circ$ $m\angle 1 + m\angle 3 = 180^\circ$	2. Definition of <u>supplementary</u> angles
3. $m\angle 1 + m\angle 2 = m\angle 1 + m\angle 3$	3. Transitive property
4. $m\angle 2 = m\angle 3$	4. Subtraction property of equality
5. $\angle 2 \cong \angle 3$	5. Definition of congruence



If two figures have equal measurements, then by the **definition of congruence**, the figures are congruent. If congruence is shown through a series of transformations, then the **definition of congruence in terms of rigid motions** was applied.

4. Discuss with your partner why the justification for step 5 is the definition of congruence. Summarize your discussion.

In step 4 we see that $\angle 2$ and $\angle 3$ have equal measures. The definition of congruence states that if two figures have equal measurements, then the figures are congruent. Therefore, we can use that definition as reason $\angle 2$ and $\angle 3$ are congruent.

5. List similarities between the paragraph proof and the two-column proof.
The paragraph proof and two-column proofs both reach the same conclusion, using the same steps and reasoning.



A **flowchart proof** shows the structure of the proof using boxes and connecting arrows. The justifications are written beside or below the box.



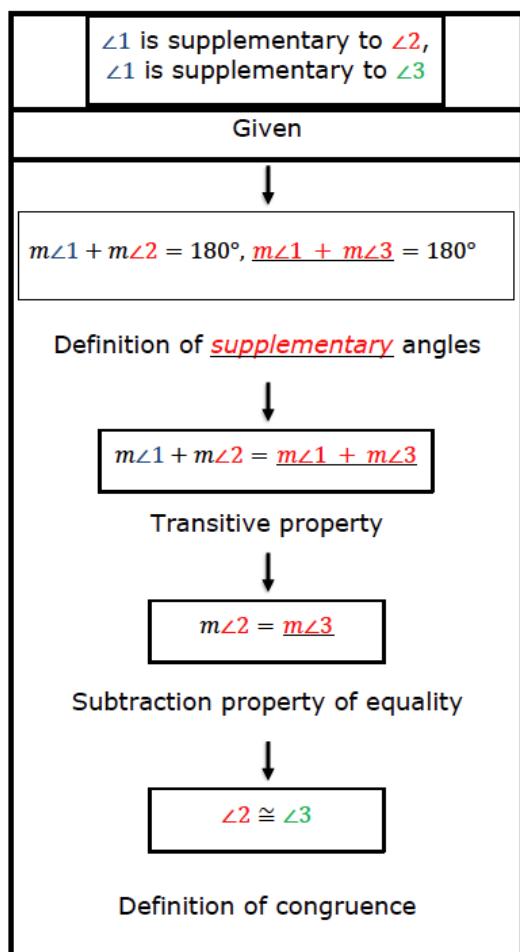
Allow students to share out their responses from their partners for other groups to hear. Often, students forget to complete the last step of a proof. Remind students that they know the conclusion of the proof so the steps should lead to that conclusion.



Have students share out their thinking to question 5. Consider leveraging an agree/disagree protocol where students raise their hand if they agree with that choice or disagree with that choice. In both the presentation of the answer and in the agree/disagree component, have students articulate why they chose that response.

2.7.2 Exploration: Proofs (continued)

6. Complete the flowchart proof.



How does this flowchart proof provide the same structure and information that the paragraph proof did?

7. Which of the three types of proofs did you find the easiest to follow?

Answers will vary. I believe the flowchart proof is the easiest to follow.

2.7.3 Guided Instruction: Flow Chart Proof

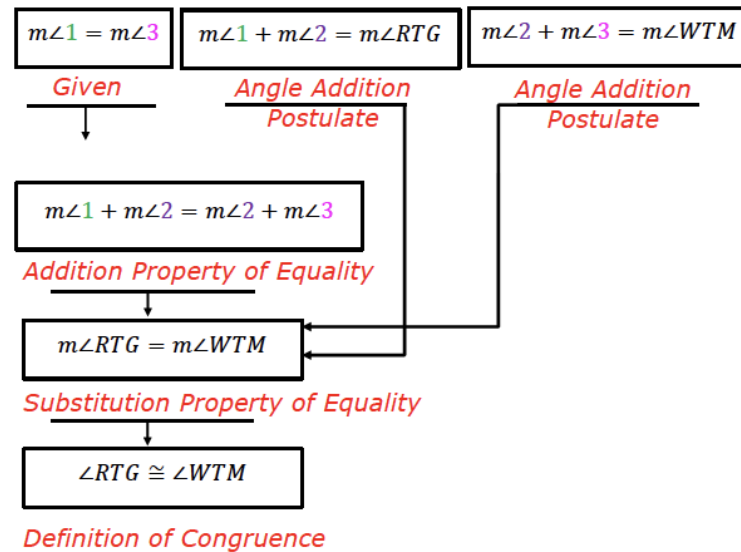
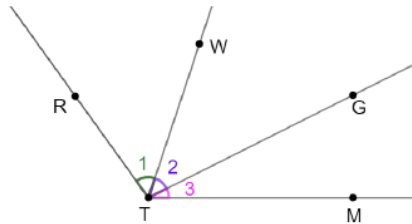
Component Overview: Teachers will guide students in completing a flowchart proof where the flow from the first statement is not a linear path in nature. Teachers should emphasize that this is a common occurrence when writing proofs.



2.7.3 Guided Instruction: Flow Chart Proof

In the previous proof, each step depended on the step before it. But this is not always true.

1. Given: $m\angle 1 = m\angle 3$
 Prove: $\angle RTG \cong \angle WTM$



Pause and ask students what the “given” and “prove” statements mean and discuss the figure shown. This is the first proof that they are seeing in this way, and not all students will recognize what this means. It is also helpful to give students a chance to do a first read of the figure so that they can process the information before moving forward with the notation and formalizing of a proof.



This proof does not have a linear flow like the previous proof. Consider asking these questions:

- How are the arrows different than in the previous flowchart proof?
- Why are three arrows drawn to the one box that states $m\angle RTG = m\angle WTM$?

2.7.4 Guided Practice: Two-Column Proof

Component Overview: Students will complete a two-column proof for the same proof in the previous component. For this activity, students can work independently or with a partner.

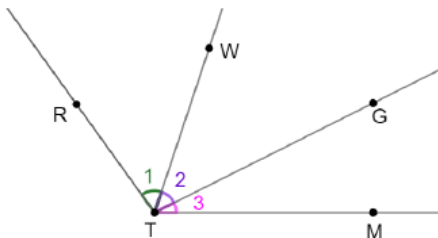


2.7.4 Guided Practice: Two-Column Proof

1. Given: $m\angle 1 = m\angle 3$

Prove:
 $\angle RTG \cong \angle WTM$

Complete the two-column proof.



Statement	Reason
1. $m\angle 1 = m\angle 3$	1. Given
2. $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$	2. Addition property of equality
3. $m\angle 1 + m\angle 2 = m\angle RTG$	3. Angle addition postulate
4. $m\angle 2 + m\angle 3 = m\angle WTM$	4. Angle addition postulate
5. $m\angle RTG = m\angle WTM$	5. Substitution property of equality
6. $\angle RTG \cong \angle WTM$	6. Definition of congruence



Note that this is the same figure as provided in the 2.7.3 Guided Instruction activity they just completed. Consider releasing students to try this on their own without any additional instruction. Many students may not notice these are the same. One of two events will occur:

- Students will not realize they are the same image and will try writing the proof again from scratch.
- Students will realize they are the same image and will practice the formatting of a two-column proof.

In both instances, students are getting additional practice with proofs in different capacities.



For an additional challenge, have students take this same proof and write it in paragraph proof form.

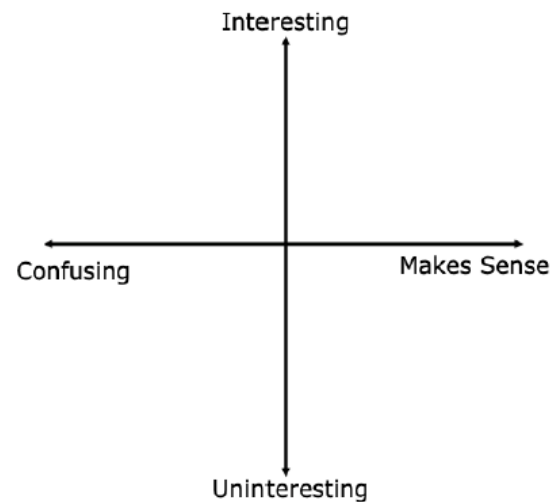
2.7.5 Wrap-Up: Reflection

Component Overview: Students will complete a self-reflection on their level of interest and understanding of this lesson. Students should complete this activity independently to provide data for differentiation in the next lesson.



2.7.5 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate how you feel about today's lesson.
Answers will vary.



Since this lesson is the first of a two-part series, it will be important to use the data from student responses in this reflection to structure the release of activities in the next lesson.